The Hedging Channel of Monetary Policy

Jin-Wook Chang[1] Antonio Falato[1] Filip Zikes[1]

[1]Federal Reserve Board

July 13-14, 2026

22nd Annual Conference of the
Asia-Pacific Association of Derivatives

Motivation + Why Derivatives Matter

- Banks have fundamentally transformed over the last two decades:
  - Loan share of total assets declined from 60% to 40%
  - Trading activities, including derivatives, have grown substantially

Motivation + Why Derivatives Matter

- Banks have fundamentally transformed over the last two decades:
  - Loan share of total assets declined from 60% to 40%
  - Trading activities, including derivatives, have grown substantially
- Are bank balance sheets still important for monetary policy?
  - Traditional view: Bank lending channel (Kashyap & Stein, 2000)
  - Skeptical view: Rise of NBFIs (Buchak et al., 2024)

Motivation + Why Derivatives Matter

- Banks have fundamentally transformed over the last two decades:
  - Loan share of total assets declined from 60% to 40%
  - Trading activities, including derivatives, have grown substantially
- Are bank balance sheets still important for monetary policy?
  - Traditional view: Bank lending channel (Kashyap & Stein, 2000)
  - Skeptical view: Rise of NBFIs (Buchak et al., 2024)
- Research Question:
  - Does monetary policy affect banks' derivatives (hedging) provision?
  - Does it depend on bank capital (balance sheets)?

Motivation + Why Derivatives Matter

- Banks have fundamentally transformed over the last two decades:
  - Loan share of total assets declined from 60% to 40%
  - Trading activities, including derivatives, have grown substantially
- Are bank balance sheets still important for monetary policy?
  - Traditional view: Bank lending channel (Kashyap & Stein, 2000)
  - Skeptical view: Rise of NBFIs (Buchak et al., 2024)
- Research Question:
  - Does monetary policy affect banks' derivatives (hedging) provision?
  - Does it depend on bank capital (balance sheets)?
- Derivatives are off-balance-sheet but subject to regulatory constraints (capital requirements through RWA, SLR, and etc.)

Motivation + Why Derivatives Matter

- Banks have fundamentally transformed over the last two decades:
  - Loan share of total assets declined from 60% to 40%
  - Trading activities, including derivatives, have grown substantially
- Are bank balance sheets still important for monetary policy?
  - Traditional view: Bank lending channel (Kashyap & Stein, 2000)
  - Skeptical view: Rise of NBFIs (Buchak et al., 2024)
- Research Question:
  - Does monetary policy affect banks' derivatives (hedging) provision?
  - Does it depend on bank capital (balance sheets)?
- Derivatives are off-balance-sheet but subject to regulatory constraints (capital requirements through RWA, SLR, and etc.)
- Real economic consequences (**hedging channel**):
  - Firms use derivatives for **hedging** interest rate, FX, commodity risks
  - **Reduced access** affects investment, precautionary savings, etc.

Preview of Main Results

- A 100 basis point increase in federal funds rate reduces:
  - Total notional amount of derivatives by approximately 8%
  - Credit Valuation Adjustment (CVA) by approximately 6%

Preview of Main Results

- A 100 basis point increase in federal funds rate reduces:
  - Total notional amount of derivatives by approximately 8%
  - Credit Valuation Adjustment (CVA) by approximately 6%

- Effect is highly heterogeneous across banks:
  - Weakly capitalized banks: 10.5% reduction in notional, 8.4% reduction in CVA
  - Well-capitalized banks: Maintain provision (little to no reduction)

Preview of Main Results

- A 100 basis point increase in federal funds rate reduces:
  - Total notional amount of derivatives by approximately 8%
  - Credit Valuation Adjustment (CVA) by approximately 6%

- Effect is highly heterogeneous across banks:
  - Weakly capitalized banks: 10.5% reduction in notional, 8.4% reduction in CVA
  - Well-capitalized banks: Maintain provision (little to no reduction)

- Operates through both intensive and extensive margins:
  - Intensive: Reduce exposure to existing counterparties
  - Extensive: Terminate relationships, curtail new formations

Preview of Main Results

- A 100 basis point increase in federal funds rate reduces:
  - Total notional amount of derivatives by approximately 8%
  - Credit Valuation Adjustment (CVA) by approximately 6%

- Effect is highly heterogeneous across banks:
  - Weakly capitalized banks: 10.5% reduction in notional, 8.4% reduction in CVA
  - Well-capitalized banks: Maintain provision (little to no reduction)

- Operates through both intensive and extensive margins:
  - Intensive: Reduce exposure to existing counterparties
  - Extensive: Terminate relationships, curtail new formations

- No evidence of demand-side effects once bank capital controlled for

Related Literature

- **Changes in bank business models:** Falato et al. (2025); Hanson et al. (2024)
  - We link this structural shift to monetary policy transmission
- **Bank lending channel & the balance-sheet debate:** Kashyap & Stein (2000); Jiménez et al. (2014); Greenwald et al. (2025); Buchak et al. (2024) vs. Stein (2024)
  - Balance sheets still transmit policy — through derivatives
- **Deposits channel & bank market power:** Drechsler et al. (2017); Egan et al. (2022); Wang et al. (2022); DeMarzo et al. (2024)
  - Embedded in our model; interacts with capital constraints
- **Intermediary balance sheets & asset prices:** Andersen et al. (2019); Fleckenstein & Longstaff (2020); Du et al. (2023)
  - Balance-sheet costs shape *provision*, not just prices
- **Hedging demand & swaps:** Froot et al. (1993); McPhail et al. (2024)
  - We identify the supply side; counterparty risk across all derivatives

Counterparty Credit Risk in Derivatives

- Banks maintain **balanced derivative positions** (no market risk)
  But balance does NOT eliminate counterparty credit risk

Counterparty Credit Risk in Derivatives

- Banks maintain **balanced derivative positions** (no market risk)
  But balance does NOT eliminate counterparty credit risk

- Example: Interest Rate Swaps
  - Bank pays fixed/receives floating from a G-SIB: $-\$10$B
  - Bank receives fixed/pays floating from a Hedge Fund: $+\$10$B
  - Market risk: Perfectly hedged (\$0 net exposure)
  - Counterparty risk: Large exposure to hedge fund (\$10B at risk)

Counterparty Credit Risk in Derivatives

- Banks maintain **balanced derivative positions** (no market risk)
  But balance does NOT eliminate counterparty credit risk

- Example: Interest Rate Swaps
  - Bank pays fixed/receives floating from a G-SIB: $-\$10$B
  - Bank receives fixed/pays floating from a Hedge Fund: $+\$10$B
  - Market risk: Perfectly hedged ($0 net exposure)
  - Counterparty risk: Large exposure to hedge fund ($10B at risk)

- Why counterparty credit risk cannot be "flat":
  - Bilateral and counterparty-specific
  - Hedging creates new counterparty exposures
  - Regulatory capital explicitly required

Data Sources

- **Main data: FR Y-14Q (Confidential)**
  - Schedule L: Counterparty-level derivative exposures
  - Schedule H.1: Corporate loan data
  - Top 10 banks (90% of total derivative assets), 2014Q4 - 2025Q3

Data Sources

- **Main data: FR Y-14Q (Confidential)**
  - Schedule L: Counterparty-level derivative exposures
  - Schedule H.1: Corporate loan data
  - Top 10 banks (90% of total derivative assets), 2014Q4 - 2025Q3

- **Key measures of derivative provision:**
  - *Total Notional Amount*: Reference amount on which payments are calculated (scale of derivative positions)
  - *Credit Valuation Adjustment (CVA)*: Market value of counterparty credit risk (expected loss from default)

$$\text{CVA}_{\text{counterparty}} = \text{LGD} \times \sum_{t=1}^{T} \text{EE}(t) \times \text{PD}(t-1,t) \times \text{DF}(t)$$

  - Also examine: Gross CE, Net CE

Data Sources

- **Main data: FR Y-14Q (Confidential)**
  - Schedule L: Counterparty-level derivative exposures
  - Schedule H.1: Corporate loan data
  - Top 10 banks (90% of total derivative assets), 2014Q4 - 2025Q3

- **Key measures of derivative provision:**
  - *Total Notional Amount*: Reference amount on which payments are calculated (scale of derivative positions)
  - *Credit Valuation Adjustment (CVA)*: Market value of counterparty credit risk (expected loss from default)

$$\mathrm{CVA}_{\mathrm{counterparty}} = \mathrm{LGD} \times \sum_{t=1}^{T} \mathrm{EE}(t) \times \mathrm{PD}(t-1,t) \times \mathrm{DF}(t)$$

  - Also examine: Gross CE, Net CE

- **Additional data:**
  - FR Y-9C: Bank balance sheet variables, capital ratios
  - DFAST: Stress test results (regulatory capital measures)
  - Monetary policy shocks (Jarociński & Karadi, 2020)

Model Overview

- Novel model of banks' optimal derivative provision:
  Extending Drechsler et al. (2017); Repullo (2020)

- Key features:
  1. Asset production (loans, securities): $f(A_{jt}, r_{ft}) \uparrow$ in A_{jt} (asset)
  2. Deposit demand (w/market power): $D(s_{jt}, r_{ft}) \downarrow$ in s_{jt} (spread)

Model Overview

- Novel model of banks' optimal derivative provision:
  Extending Drechsler et al. (2017); Repullo (2020)

- Key features:
  1. Asset production (loans, securities): $f(A_{jt}, r_{ft}) \uparrow$ in A_{jt} (asset)
  2. Deposit demand (w/market power): $D(s_{jt}, r_{ft}) \downarrow$ in s_{jt} (spread)
  3. Derivative revenue: $H(C_{jt}, r_{ft}) \uparrow$ in C_{jt} (counterparty exposure)
  4. Regulatory capital constraint: $\frac{E_{jt}}{A_{jt}+\gamma C_{jt}} \geq \kappa$

  + concavity assumptions

Model Overview

- Novel model of banks' optimal derivative provision:
  Extending Drechsler et al. (2017); Repullo (2020)

- Key features:
  1. Asset production (loans, securities): $f(A_{jt}, r_{ft}) \uparrow$ in A_{jt} (asset)
  2. Deposit demand (w/market power): $D(s_{jt}, r_{ft}) \downarrow$ in s_{jt} (spread)
  3. Derivative revenue: $H(C_{jt}, r_{ft}) \uparrow$ in C_{jt} (counterparty exposure)
  4. Regulatory capital constraint: $\frac{E_{jt}}{A_{jt}+\gamma C_{jt}} \geq \kappa$

  + concavity assumptions

- Bank j maximizes:

$$\max_{s_{jt}, C_{jt}} \quad f(A_{jt}, r_{ft}) - (r_{ft} - s_{jt})D(s_{jt}, r_{ft}) + H(C_{jt}, r_{ft})$$

$$\text{s.t.} \quad A_{jt} = E_{jt} + D(s_{jt}, r_{ft})$$

$$E_{jt} \geq \kappa(A_{jt} + \gamma C_{jt})$$

Model Overview

- Novel model of banks' optimal derivative provision:
  Extending Drechsler et al. (2017); Repullo (2020)

- Key features:
  1. Asset production (loans, securities): $f(A_{jt}, r_{ft}) \uparrow$ in A_{jt} (asset)
  2. Deposit demand (w/market power): $D(s_{jt}, r_{ft}) \downarrow$ in s_{jt} (spread)
  3. Derivative revenue: $H(C_{jt}, r_{ft}) \uparrow$ in C_{jt} (counterparty exposure)
  4. Regulatory capital constraint: $\frac{E_{jt}}{A_{jt}+\gamma C_{jt}} \geq \kappa$

  \+ concavity assumptions

- Bank j maximizes:

$$
\begin{aligned}
\max_{s_{jt}, C_{jt}} \quad & f(A_{jt}, r_{ft}) - (r_{ft} - s_{jt}) D(s_{jt}, r_{ft}) + H(C_{jt}, r_{ft}) \\
\text{s.t.} \quad & A_{jt} = E_{jt} + D(s_{jt}, r_{ft}) \\
& E_{jt} \geq \kappa(A_{jt} + \gamma C_{jt})
\end{aligned}
$$

- $s_{jt} \uparrow \implies D \downarrow \implies A \downarrow \implies f' \uparrow \implies C_{jt} \downarrow$

Key Mechanisms

- **Balance sheet cost channel:**
  - Derivatives require balance sheet space (via γ)
  - Trade-off: Allocate constrained capital to loans vs. derivatives
  - At optimum: Marginal profit from lending = Marginal revenue from derivatives

Key Mechanisms

- **Balance sheet cost channel:**
  - Derivatives require balance sheet space (via γ)
  - Trade-off: Allocate constrained capital to loans vs. derivatives
  - At optimum: Marginal profit from lending = Marginal revenue from derivatives

- **Effect of rate increase:**
  - Deposit supply increases (higher returns attract depositors)
  - Banks exploit market power: Increase deposit spreads
  - Higher spreads $\Rightarrow$ Lower deposits $\Rightarrow$ Lower assets
  - Marginal profit from lending increases
  - Relative value of balance sheet space increases
  - $\Rightarrow$ Reduce derivative provision

Key Mechanisms

- **Balance sheet cost channel:**
  - Derivatives require balance sheet space (via γ)
  - Trade-off: Allocate constrained capital to loans vs. derivatives
  - At optimum: Marginal profit from lending = Marginal revenue from derivatives
- **Effect of rate increase:**
  - Deposit supply increases (higher returns attract depositors)
  - Banks exploit market power: Increase deposit spreads
  - Higher spreads $\Rightarrow$ Lower deposits $\Rightarrow$ Lower assets
  - Marginal profit from lending increases
  - Relative value of balance sheet space increases
  - $\Rightarrow$ Reduce derivative provision
- **Role of capital:**
  - More capital $\Rightarrow$ more balance sheet space (shadow value of space $\downarrow$) $\Rightarrow$ lending vs derivatives trade-off is slack
  - Tightening forces less reallocation out of derivatives

Baseline Specification

- Bank i - Counterparty j - Quarter t panel regression:

$$\tilde{y}_{ijt} = \mu_i + \mu_j + \beta_1 \Delta r_t + \beta_2 E_{it} + \beta_3 (\Delta r_t \times E_{it}) + \epsilon_{ijt}$$

- Where:
  - $\tilde{y}_{ijt}$: Symmetric change in Total Notional or CVA
  - Δr_t: Change in effective federal funds rate
  - E_{it}: Bank capital proxy (Tier 1 Capital / RWA)
  - μ_i, μ_j: Bank and counterparty fixed effects

Baseline Specification

- Bank i - Counterparty j - Quarter t panel regression:

$$\tilde{y}_{ijt} = \mu_i + \mu_j + \beta_1 \Delta r_t + \beta_2 E_{it} + \beta_3 (\Delta r_t \times E_{it}) + \epsilon_{ijt}$$

- Where:
  - $\tilde{y}_{ijt}$: Symmetric change in Total Notional or CVA
  - Δr_t: Change in effective federal funds rate
  - E_{it}: Bank capital proxy (Tier 1 Capital / RWA)
  - μ_i, μ_j: Bank and counterparty fixed effects

- Identification:
  - Fixed effects control for time-invariant bank/counterparty heterogeneity
  - Multiple robustness checks and alternative specifications
  - Address endogeneity with stress test capital measures

Main Results: Baseline

	Without Capital		With Capital	
	Notional	CVA	Notional	CVA
Rate Change	-0.082^{***}	-0.060^{***}	-0.422^{***}	-0.419^{***}
	(-9.44)	(-6.81)	(-6.41)	(-6.14)
Bank Capital Proxy			2.004^{**}	-0.400
			(2.56)	(-0.68)
Rate Change $\times$ Capital			2.423^{***}	2.557^{***}
			(5.23)	(5.37)
R-squared	0.734	0.599	0.734	0.599
Observations	114,431	149,307	113,665	148,541

- 100 bps rate increase: 8.2% $\downarrow$ in notional, 6.0% $\downarrow$ in CVA
- **Strong heterogeneity**: Positive interaction with bank capital
- Results remain robust with counterparty $\times$ time FEs

Economic Magnitude

- Consider differential effect across capital distribution:

- **Total Notional Amount:**
  - Bank at 25th percentile capital (13.1%):
    $-0.422 + 2.423 \times 0.131 = -0.105 \Rightarrow$ 10.5% reduction
  - Bank at 75th percentile capital (17.0%):
    $-0.422 + 2.423 \times 0.170 = -0.010 \Rightarrow$ 1.0% reduction
  - **Difference: 9.5 percentage points**

Economic Magnitude

- Consider differential effect across capital distribution:

- **Total Notional Amount:**
    - Bank at 25th percentile capital (13.1%):
      $-0.422 + 2.423 \times 0.131 = -0.105 \Rightarrow$ 10.5% reduction
    - Bank at 75th percentile capital (17.0%):
      $-0.422 + 2.423 \times 0.170 = -0.010 \Rightarrow$ 1.0% reduction
    - **Difference: 9.5 percentage points**

- **CVA:**
    - Bank at 25th percentile:
      $-0.419 + 2.557 \times 0.131 = -0.084 \Rightarrow$ 8.4% reduction
    - Bank at 75th percentile:
      $-0.419 + 2.557 \times 0.170 = +0.016 \Rightarrow$ 1.6% increase
    - **Difference: 10.0 percentage points**

Addressing Endogeneity: Stress Test Capital

- **Concern:** Bank capital may be correlated with derivative decisions

Addressing Endogeneity: Stress Test Capital

- **Concern:** Bank capital may be correlated with derivative decisions
- **Solution:** Stress test regulatory capital (Berrospide & Edge, 2024)
  - Fed Model Capital: Determined by regulators, not banks
  - Capital Gap: Required minus actual capital
  - Plausibly exogenous to individual derivative decisions

Addressing Endogeneity: Stress Test Capital

- **Concern:** Bank capital may be correlated with derivative decisions
- **Solution:** Stress test regulatory capital (Berrospide & Edge, 2024)
  - Fed Model Capital: Determined by regulators, not banks
  - Capital Gap: Required minus actual capital
  - Plausibly exogenous to individual derivative decisions
- **Results remain robust:**

	Notional	CVA
Rate Change	-0.378^{***}	-0.371^{***}
	(-3.75)	(-3.59)
Fed Model Capital, Severe	0.0145^{*}	-0.0167^{***}
	(1.83)	(-2.85)
Rate Change $\times$ Fed Model Capital	0.0297^{***}	0.0324^{***}
	(2.88)	(3.08)
R-squared	0.760	0.613
Observations	20,655	27,231

Conclusion and Policy Implications

1. **A new monetary transmission channel through derivatives**
   - 100 bps rate increase: 8-10% reduction in derivative provision (weakly capitalized banks)
   - Well-capitalized banks maintain provision
   - Balance sheets still matter, but through evolved mechanisms
2. **Bank balance sheet cost channel**
   - Post-GFC regulations made derivatives capital-intensive
   - Banks reallocate constrained capital between lending and derivatives
   - Reduced hedging access may amplify monetary tightening effects
3. **Policy and regulatory trade-offs**
   - Capital requirements enhance stability but affect transmission
   - Tightening increases market concentration (largest banks gain share)
   - One-size-fits-all approach may be suboptimal

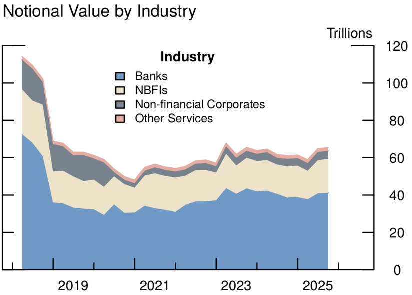
Appendix

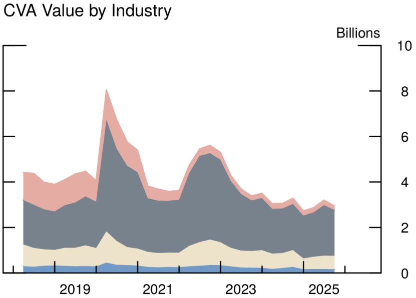
Summary Statistics

	Mean	SD	25%	Median	75%	Obs
Panel A: Bank-Counterparty-Quarter						
Δ Total Notional	0.027	1.383	-1.338	0.038	1.388	207,152
Δ CVA	0.070	1.106	-0.836	0.096	1.001	207,152
Credit Rating	4.117	1.435	3	4	5	207,152
Panel B: Bank-Quarter						
Bank Capital Proxy	0.155	0.036	0.131	0.141	0.170	360
Total Assets (Trillions)	1.539	0.964	0.832	1.704	2.233	360
Panel C: Quarter						
Rate Change	0.092	0.422	0	0.020	0.150	45

- Symmetric change: $\tilde{y}_{ijt} = \frac{y_{ijt} - y_{ij,t-1}}{0.5(y_{ijt} + y_{ij,t-1})}$ (Davis & Haltiwanger, 1992)
  - Bounded: $-2 \leq \tilde{y}_{ijt} \leq 2$
  - Approximates percentage change for moderate changes
  - Handles entries $(0 \rightarrow +)$ and exits $(+ \rightarrow 0)$
- Bank Capital Proxy: Tier 1 Capital / Risk-Weighted Assets

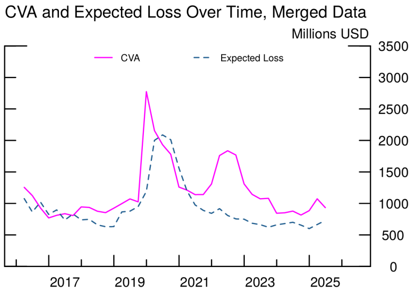
Time Series of Derivative Exposures

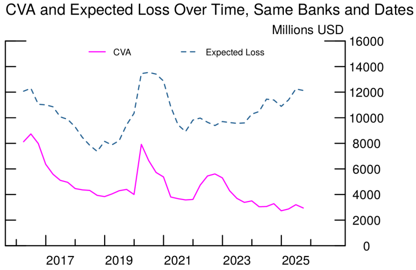




- Exposures to financials are mitigated by netting and collateral
- Non-financial corporates account for the largest proportion of CVA

CVA vs Expected Losses on Loans





- CVA and expected losses on loans (LGD $\times$ PD) are comparable
- They may comove for the same set of counterparties at the same quarter for certain events (e.g., COVID-19)
- Comovement becomes significantly weaker for full sample

Model Predictions

Proposition (Capital Constraint Tightness)

A tighter capital constraint (higher κ) leads to:

- *Higher deposit spreads s^*_{jt}*
- *Lower derivative provision C^*_{jt}*

Model Predictions

Proposition (Capital Constraint Tightness)

A tighter capital constraint (higher κ) leads to:

- *Higher deposit spreads s^*_{jt}*
- *Lower derivative provision C^*_{jt}*

Proposition (Bank Capital)

An increase in bank capital E_{jt}:

- *Always increases derivative provision C^*_{jt}*
- *Effect on deposit spreads depends on relative concavity*

Interest Rate Sensitivity

Proposition (Federal Funds Rate)

Effect of r_{ft} increase depends on derivative demand sensitivity ξ:

- *If ξ is low: Derivative provision decreases*
- *If ξ is high: Derivative provision increases*

Interest Rate Sensitivity

Proposition (Federal Funds Rate)

Effect of r_{ft} increase depends on derivative demand sensitivity ξ:

- *If ξ is low: Derivative provision decreases*
- *If ξ is high: Derivative provision increases*

Proposition (Interest Rate Elasticity)

Suppose derivative provision decreases with r_{ft}. Then:

- *Interest rate elasticity becomes* ***less elastic*** *when E_{jt} increases*
- *Banks with more capital reduce derivative provision less*

Event Study: 2022-2023 Tightening

	Without Capital		With Capital	
	Notional	CVA	Notional	CVA
Rate Hike	-0.123^{***}	-0.126^{***}	-0.553^{***}	-0.623^{***}
	(-7.19)	(-7.35)	(-4.31)	(-4.79)
Bank Capital Proxy			1.244	-1.121^{*}
			(1.61)	(-1.88)
Rate Hike × Capital			3.002^{***}	3.473^{***}
			(3.33)	(3.84)
R-squared	0.734	0.600	0.734	0.599
Observations	114,431	149,307	113,665	148,541

- Rate Hike = 1 for 2022Q2 onwards
- Even stronger effects during aggressive tightening
- Larger interaction coefficients: Capital buffers more important during tightening

Monetary Policy Shocks

	Without Capital		With Capital	
	Notional	CVA	Notional	CVA
PC1 MP Shock	-0.040^{***}	-0.053^{***}	-0.218^{***}	-0.396^{***}
	(-6.86)	(-9.20)	(-4.74)	(-8.68)
Bank Capital Proxy			1.338^{*}	-0.584
			(1.71)	(-1.00)
PC1 MP Shock $\times$ Capital			1.264^{***}	2.410^{***}
			(4.02)	(7.84)
R-squared	0.734	0.599	0.733	0.599
Observations	113,665	148,541	113,665	148,541

- Use high-frequency identified shocks (Jarociński & Karadi, 2020)
- Weighted by distance from quarter-end (Ottonello & Winberry, 2020)
- Results consistent with baseline
- Addresses reverse causality concerns

Supply vs Demand: Counterparty Balance Sheets

- **Question:** Are results driven by supply (banks) or demand (firms)?

Supply vs Demand: Counterparty Balance Sheets

- **Question:** Are results driven by supply (banks) or demand (firms)?
- Merge with loan data to examine counterparty firm characteristics:
  - Committed Exposure (loans)
  - Total Assets
  - Total Debt
  - Cash Holdings

Supply vs Demand: Counterparty Balance Sheets

- **Question:** Are results driven by supply (banks) or demand (firms)?

- Merge with loan data to examine counterparty firm characteristics:
  - Committed Exposure (loans)
  - Total Assets
  - Total Debt
  - Cash Holdings

- **Finding:** Rate changes affect firm variables...
  - ...but effects disappear once we control for bank capital

Supply vs Demand: Counterparty Balance Sheets

- **Question:** Are results driven by supply (banks) or demand (firms)?
- Merge with loan data to examine counterparty firm characteristics:
  - Committed Exposure (loans)
  - Total Assets
  - Total Debt
  - Cash Holdings
- **Finding:** Rate changes affect firm variables...
  - ...but effects disappear once we control for bank capital
- **Interpretation:**
  - No evidence of demand-side effects conditional on bank capital
  - Results driven by bank supply-side responses
  - Consistent with balance sheet cost channel

Credit Risk Management

	Notional	CVA
Rate Change	-0.351^{***}	-0.338^{***}
	(-5.01)	(-4.62)
Credit Rating	-0.031^{***}	0.058^{***}
	(-3.25)	(7.34)
Hedge	0.046	0.130^{***}
	(1.41)	(4.55)
Rate Change × Capital	2.448^{***}	2.369^{***}
	(5.32)	(4.96)
Rate Change × Credit Rating	-0.018^{***}	-0.015^{**}
	(-2.94)	(-2.45)
Rate Change × Hedge	-0.029	0.058^{**}
	(-1.04)	(1.98)
R-squared	0.732	0.598
Observations	118,276	152,215

- **Hedge** $= 1$ if bank holds single-name credit hedge (e.g., CDS)
- **Credit Rating**: Lower value = better credit quality
- Hedge raises CVA (selection effect) and dampens rate impact
- Better rating ⇒ more notional but less CVA

Controlling for Lending

	Rate Only		+ Capital Controls	
	Notional	CVA	Notional	CVA
Rate Change	-0.129^{***}	-0.062^{***}	-0.711^{***}	-0.472^{**}
	(-6.98)	(-2.84)	(-4.52)	(-2.55)
Bank Capital Proxy (own)			1.243	-2.466^{*}
			(0.75)	(-1.78)
Rate Change × Capital (own)			4.130^{***}	2.726^{**}
			(3.80)	(2.32)
H Bank Capital Proxy			0.660	-1.022
			(0.63)	(-0.81)
Rate Change × H Bank Capital			0.134	0.256
			(0.19)	(0.31)
R-squared	0.454	0.326	0.455	0.326
Observations	27,264	31,639	27,252	31,627

- **Subsample:** Counterparties with *both* derivative and C&I lending
- **H Bank Capital Proxy**: Weighted-avg Tier 1/RWA across *all* lending banks
- Own-bank capital interaction *70% larger* than baseline
- Cross-bank lending strength (H Bank Capital) does *not* matter ⇒ provision determined by *specific* dealer's capital

Intensive vs Extensive Margins

- Do banks adjust through:
    - **Intensive margin:** Change exposure to existing counterparties?
    - **Extensive margin:** Form/terminate relationships?

Intensive vs Extensive Margins

- Do banks adjust through:
  - **Intensive margin:** Change exposure to existing counterparties?
  - **Extensive margin:** Form/terminate relationships?

- Extend sample to include relationship entries/exits

	Notional	CVA
Intensive Only (Baseline)		
Rate Change × Capital	2.423^{***}	2.557^{***}
Intensive + Extensive		
Rate Change × Capital	3.576^{***}	5.767^{***}

- Extensive margin amplifies effects
- Less-capitalized banks terminate relationships during tightening
- Well-capitalized banks gain market share